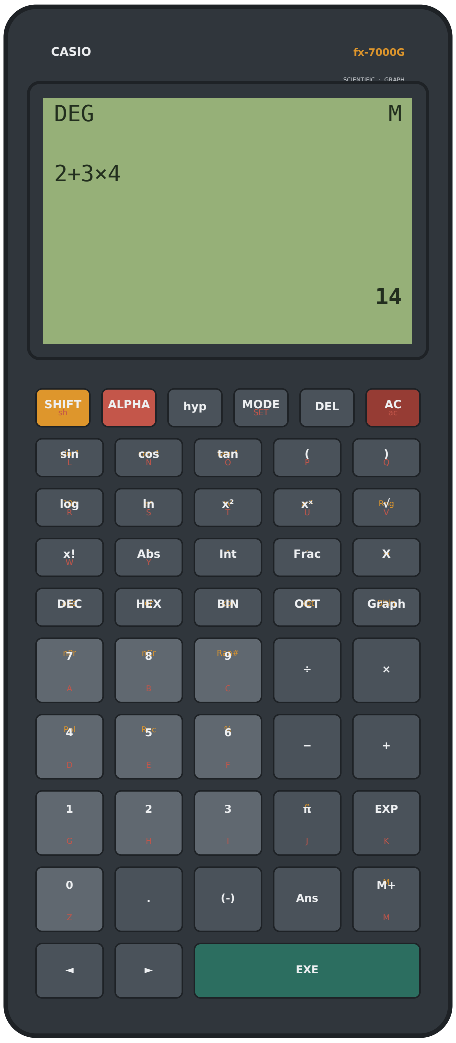
# CASIO

fx-7000G  Scientific & Graphing Calculator

# Owner's Manual

Operation guide · Worked examples · Key reference




A faithful recreation of the 1985 CASIO fx-7000G — Kotlin + Jetpack Compose edition.
Dot-matrix 96 × 64 LCD · DEG/RAD/GRA · BASE-n · function graphing.

# Preface

Congratulations on your fx-7000G. This calculator revives the spirit of the machine that, in 1985, became the world's first graphing scientific calculator — now rebuilt as a modern Android application with a genuine dot-matrix display, a full scientific engine and function graphing.

This manual is your complete guide. It explains every key, every operating mode and every function, and it is filled with worked examples set out in the classic “Operation / Display” style: read the key sequence on the left, and see exactly what appears on the LCD on the right. Follow along on your own calculator and you will master it in an afternoon.

The examples in this book are not decorative — each one is a real calculation the engine performs, with the exact result the display produces. Where a key has more than one meaning, the manual shows how the SHIFT, ALPHA and hyp prefixes unlock its second and third functions.

### How to use this manual

- If you have never used the calculator before, read Chapter 1 (General Guide) first — it explains the display and the prefix keys that everything else depends on.
- For everyday arithmetic and editing, see Chapter 2.
- Use the Table of Contents (overleaf) to jump to a topic, and the alphabetical Index at the back to look up a specific function or key.
- Throughout, a key on the keypad is drawn as a chip, e.g. the EXE key, and multi-key operations are shown as a sequence.

> **Reading the symbols:** In this manual the display glyphs are printed exactly as they appear on the calculator: × and ÷ for multiply and divide, − for subtract/negative, √ for square root, π for pi and → for the variable-store arrow.

# Contents

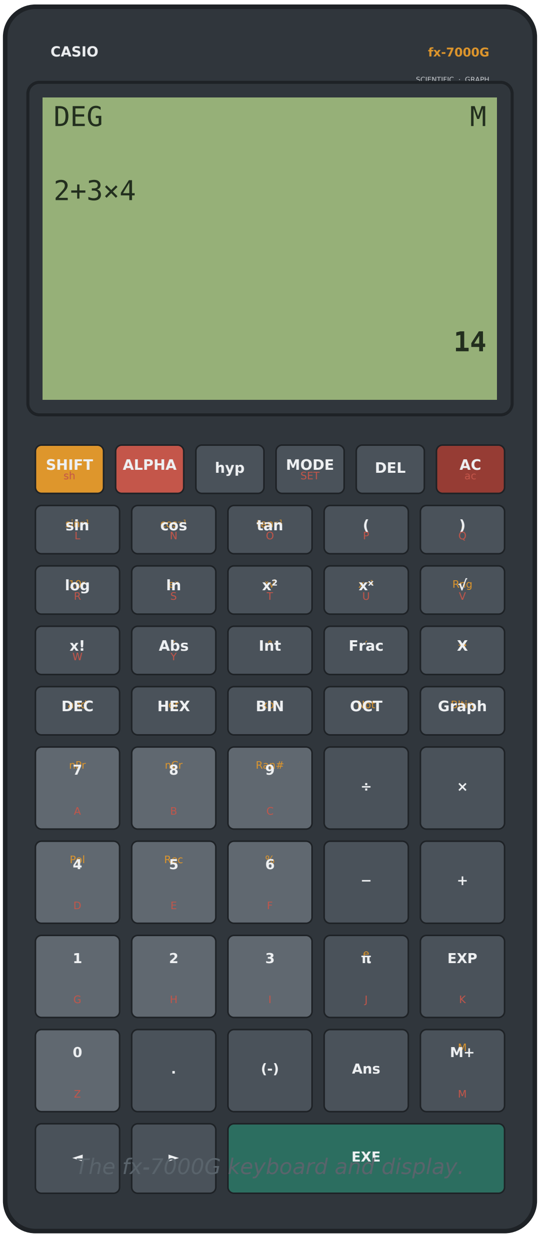
**CHAPTER 1**

# General Guide

This chapter introduces the parts of the calculator you will use in every calculation: the display, the keyboard, and the three prefix keys that give each button more than one function.

## 1-1 The keyboard

The keyboard is arranged in ten rows. The upper five rows are the slim function keys (scientific functions, modes and base conversion); the lower rows carry the numbers, the four arithmetic operators and the large EXE key that evaluates an entry. Most keys print a small legend above and below the main label: the upper (orange) legend is reached with SHIFT, the lower (red) legend with ALPHA.




The fx-7000G keyboard and display.

## 1-2 The prefix keys: SHIFT, ALPHA and hyp

Three keys do not enter anything by themselves. Instead they change the meaning of the next key you press. They are one-shot: after the next key they switch off automatically. While a prefix is active a small indicator (S, A or h) appears at the right of the status line.

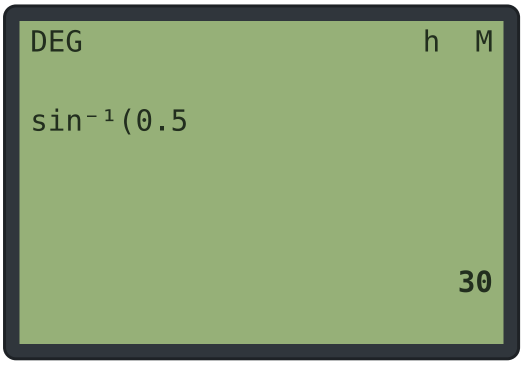
- SHIFT selects the function printed in orange above a key. Example: SHIFT then the sin key gives $\sin^{-1}$ (arc-sine).
- ALPHA selects the red letter printed below a key, so you can enter the variables A through Z.
- hyp turns the sin / cos / tan keys into the hyperbolic functions sinh, cosh and tanh. Press SHIFT before hyp to reach the inverse hyperbolics $\sinh^{-1}$, $\cosh^{-1}$ and $\tanh^{-1}$.

> **Note:** SHIFT and ALPHA are mutually exclusive — turning one on turns the other off. Pressing a prefix a second time cancels it.

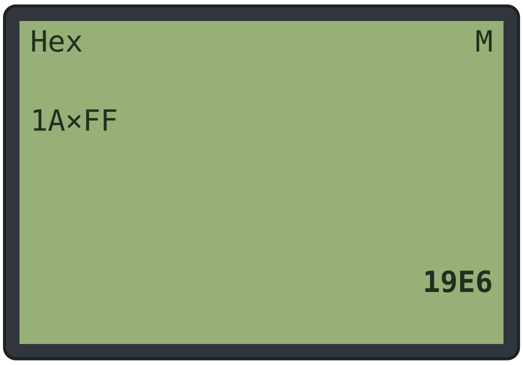
## 1-3 The display

The screen is a dot-matrix liquid-crystal display of 96 × 64 pixels. Text is drawn on a 5 × 7 pixel font inside 6 × 8 cells, giving 16 characters across and up to 8 rows. In calculation the layout is:

- Status line (top): the current mode at the left (DEG, RAD, GRA, or the base name Hex/Oct/Bin/Dec), and at the right the S / A / h prefix flags and the M memory flag.
- Entry line: the expression you are typing, over two rows if needed (up to 32 characters), with the cursor kept in view.
- Result line: the answer, right-aligned, shown after you press EXE.



*Status flags, an entry line and a right-aligned result.*



*BASE-n mode: the active base replaces the angle unit.*

## 1-4 Entering and evaluating

Type an expression using the number and function keys, then press EXE to evaluate it. The result appears on the result line and is remembered as Ans (see Chapter 2). To correct a mistake use DEL to erase the character before the cursor, or AC to clear the whole entry.

**2** → **+** → **3** → **×** → **4** → **EXE**

The sequence above computes 2 + 3 × 4. Because × binds tighter than +, the answer is 14 (not 20).

## 1-5 Appearance and screen fit

Two controls on the calculator's face let you change how the app looks and how it fits your device. Both remember their setting and survive restarting the app.

- Theme — press and hold the CASIO fx-7000G name at the top of the body to switch between the default dark-grey theme and a classic brushed-aluminium skin. The classic skin uses cream function keys with black lettering and black number keys with white lettering, while the AC, EXE, SHIFT and ALPHA keys keep their signature colours.
- Screen fit — tap the red GRAPHICS badge to the right of the name to change how much of the screen the calculator uses. When it is on, the layout stays clear of the status/notification bar, the navigation bar and any camera cut-out; when it is off, the calculator stretches edge-to-edge to fill the whole display.

> **Note:** Both toggles are cosmetic — they never change a calculation or the value stored in memory.

CHAPTER 2

# Basic Calculations

## 2-1 Arithmetic and precedence

The four operators +, −, × and ÷ work as you expect. Multiplication and division are evaluated before addition and subtraction; use parentheses to override the order. The closing parenthesis may be omitted at the end of an expression — the calculator supplies it automatically when you press EXE.

**Example 1 — order of operations**

| Operation | Display |
|---|---|
| `2+3×4` EXE | **14** |
| `(2+3)×4` EXE | **20** |
| `2×(3+4` EXE | **14** |

*The last line shows the optional closing parenthesis: 2×(3+4 is read as 2×(3+4).*

## 2-2 Negative numbers

Use the (-) key to enter a negative value, and the − key for subtraction. Both print the same minus glyph on the display; the engine understands a leading minus as negation.

**Example 2 — signed values**

| Operation | Display |
|---|---|
| `(-)5+8` EXE | **3** |
| `8−(-)5` EXE | **13** |

## 2-3 Implicit multiplication

You may leave out the × sign in front of a constant, a variable, an opening parenthesis or a function. The calculator inserts the multiplication for you. This makes expressions read like ordinary mathematics.

**Example 3 — omitting the multiply sign**

| Operation | Display |
|---|---|
| `2π` EXE | **6.283185307** |
| `2(3+2)` EXE | **10** |
| `(2)(3)` EXE | **6** |

## 2-4 Answer memory (Ans)

The result of every calculation is stored in Ans. Press the Ans key to insert it into a new expression. As a shortcut, if you press an operator immediately after EXE, the calculator begins the new entry with Ans automatically — so you can chain calculations.

| Example 4 — chaining with Ans | |
|---|---|
| **Operation** | **Display** |
| 150×1.05 EXE | **157.5** |
| + (auto Ans) | **Ans+** |
| 10 EXE | **167.5** |
| Ans÷2 EXE | **83.75** |

## 2-5 Editing an entry

Move the cursor with the ◄ and ► keys. Pressing an arrow just after EXE replays the previous expression so you can edit and re-run it. DEL deletes the item to the left of the cursor — a whole function name such as sin( or Ans is removed in a single press. AC clears everything and also erases any graph.

- ◄ / ► move the cursor (or, right after EXE, recall the last entry for editing).
- DEL erase one character or one whole token to the left.
- AC clear the entry, the result and any graph curves.

CHAPTER 3

# Modes and Display Formats

## 3-1 Angle unit — DEG / RAD / GRA

The MODE key cycles the angle unit used by the trigonometric functions: DEG → RAD → GRA → DEG. The active unit is always shown at the left of the status line. In degrees a right angle is 90, in radians π/2, and in grades 100.

**Example 5 — the same angle in three units**

| Operation | Display |
|---|---|
| `[DEG]  sin30  EXE` | **0.5** |
| `[RAD]  sin(π÷2  EXE` | **1** |
| `[GRA]  sin100  EXE` | **1** |

## 3-2 Display format — Norm / Fix / Sci

Press SHIFT then MODE (the SET function) to open the setup menu. From here you choose the angle unit or the number-display format. The menu offers:

- 1 Deg   2 Rad   3 Gra — set the angle unit directly.
- 4 Fix — then a digit 0–9 fixes that many decimal places.
- 5 Sci — then a digit 1–9 sets the number of significant figures (0 means 10).
- 6 Norm — return to the standard 10-significant-digit display.

In Norm the calculator shows up to 10 significant digits, switching to exponential notation (mantissa E exponent) for magnitudes of $10^{10}$ or greater, or smaller than $10^{-19}$. Fix rounds to a fixed number of decimals; Sci always uses exponential form.

**Example 6 — the same value in three formats**

| Operation | Display |
|---|---|
| `[Norm]  2÷3  EXE` | **0.6666666667** |
| `[Fix 4]  2÷3  EXE` | **0.6667** |
| `[Sci 3]  2÷3  EXE` | **6.67E-1** |

*Set the format from SHIFT → MODE (SET) before evaluating.*

> **Opening SET:** SET is reached with SHIFT → MODE. After choosing 4 Fix or 5 Sci the display prompts for the digit count — just press a number key.

CHAPTER 4

# Scientific Function Calculations

## 4-1 Trigonometric and inverse functions

The sin, cos and tan keys open a function with a left parenthesis, e.g. sin(. They use the current angle unit. Their SHIFT functions $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ are the inverse (arc) functions.

| Example 7 — trig and arc-trig (DEG) | |
|---|---|
| **Operation** | **Display** |
| cos60 EXE | **0.5** |
| tan45 EXE | **1** |
| $\sin^{-1}$(0.5 EXE | **30** |

## 4-2 Hyperbolic functions (hyp)

Press hyp before sin, cos or tan to get sinh, cosh and tanh. Press SHIFT then hyp then the key for the inverse hyperbolic functions.

**hyp** → **sin** → **1** → **EXE**

| Example 8 — hyperbolic | |
|---|---|
| **Operation** | **Display** |
| hyp sin1 EXE | **1.175201194** |
| hyp cos0 EXE | **1** |
| SHIFT hyp tan(0.5 EXE | **0.5493061443** |

## 4-3 Logarithms and exponentials

log is the common (base-10) logarithm and ln is the natural logarithm. Their SHIFT functions are the antilogarithms: $10^x$ and $e^x$, which begin the power expressions 10^( and e^(.

| Example 9 — logs and powers of the base | |
|---|---|
| **Operation** | **Display** |
| log100 EXE | **2** |
| ln(e EXE | **1** |
| $10^x$3 EXE | **1000** |
| $e^x$1 EXE | **2.718281828** |

## 4-4 Powers, roots and reciprocals

The calculator offers $x^2$ (square), $x^x$ (general power, the ^ key), √ (square root), $^x\sqrt{}$ (the index-th root, SHIFT of $x^2$), and $x^{-1}$ (reciprocal, SHIFT of the power key). Powers are evaluated right-to-left, so 2^3^2 is 2^(3^2) = 512.

| Example 10 — powers and roots | |
|---|---|
| **Operation** | **Display** |
| $5^{2}$ EXE | 25 |
| 2^10 EXE | 1024 |
| 2^3^2 EXE | 512 |
| √(2 EXE | 1.414213562 |
| $3^{x}\sqrt{}8$ EXE | 2 |
| $4^{-1}$ EXE | 0.25 |

*$3^{x}\sqrt{}8$ is the cube root of 8; the index is entered before the $^{x}\sqrt{}$ symbol.*

## 4-5 Factorial, percent, nPr and nCr

x! gives the factorial of a whole number (0 to 69). % divides by 100. nPr and nCr (SHIFT of 7 and 8) are the permutation and combination counts; they bind more tightly than + and − but looser than × and ÷.

| Example 11 — counting | |
|---|---|
| **Operation** | **Display** |
| 5! EXE | 120 |
| 50% EXE | 0.5 |
| 5nPr2 EXE | 20 |
| 5nCr2 EXE | 10 |

## 4-6 Abs, Int and Frac

Abs( returns the absolute value. Int( truncates toward zero to the integer part, and Frac( returns the fractional part (the value minus its integer part).

| Example 12 — parts of a number | |
|---|---|
| **Operation** | **Display** |
| Abs((-)7 EXE | 7 |
| Int(3.7 EXE | 3 |
| Frac(3.7 EXE | 0.7 |
| Int((-)3.7 EXE | -3 |

## 4-7 Coordinate conversion — Pol and Rec

Pol(x, y) converts rectangular coordinates to polar: it returns the radius r and stores the angle Π in variable J (with r also placed in I). Rec(r, θ) does the reverse, returning x and storing y in J. The comma is the SHIFT of the ) key. The angle uses the current unit.

| Example 13 — rectangular → polar (DEG) | |
|---|---|
| **Operation** | **Display** |
| Pol(3,4 EXE | 5 |
| J EXE | 53.13010235 |
| Rec(5,53.13 EXE | 3 |

*After Pol, recall the angle by pressing ALPHA then J. Results I and J are ordinary variables.*

## 4-8 Sexagesimal (degrees ° minutes ′ seconds ″)

Enter an angle in degrees-minutes-seconds using the ° (SHIFT Int), ′ (SHIFT Frac) and ″ (SHIFT Abs) marks. The calculator converts the entry to decimal degrees.

**Example 14 — DMS to decimal degrees**

| Operation | Display |
|---|---|
| 30°30′ EXE | **30.5** |
| 1°30′36″ EXE | **1.51** |

CHAPTER 5

# Memory and Variables

## 5-1 Variables A through Z

The calculator has 26 lettered memories, A to Z. Enter a letter with the ALPHA prefix. Store a value by following it with the → arrow (SHIFT of the X key) and a letter: value → A. Recall a variable simply by using its letter in an expression; an unset variable reads as 0.

**Example 15 — storing and recalling**

| Operation | Display |
|---|---|
| 5→A EXE | **5** |
| A×2 EXE | **10** |
| $A^2$+1 EXE | **26** |

*Type A with ALPHA → (the 7 key's red legend), and → with SHIFT → X.*

## 5-2 Independent memory — M+, M− and M

The M+ key adds the current value to the independent memory M; SHIFT M+ (the M− legend) subtracts it instead. Recall M with ALPHA M+ (the red M legend). Whenever M holds a non-zero value the M flag is shown on the status line. Clear it by storing 0 into M (0 → M).

**Example 16 — accumulating a running total**

| Operation | Display |
|---|---|
| 250 M+ | **250** |
| 120 M+ | **120** |
| 30 SHIFT M+ | **30** |
| ALPHA M EXE | **340** |

*M+ adds and SHIFT M+ (M−) subtracts; recall the total with ALPHA M+.*

CHAPTER 6

# BASE-n Calculations

## 6-1 Decimal, hexadecimal, binary and octal

The DEC, HEX, BIN and OCT keys switch the calculator into integer base-n mode and convert the current value to that base. The active base (Dec, Hex, Bin or Oct) is shown on the status line in place of the angle unit. In a non-decimal base the keypad accepts only the digits valid for that base — hexadecimal digits A–F are the ALPHA letters. Values are whole numbers only.

**Example 17 — converting between bases**

| Operation | Display |
|---|---|
| [DEC] 255 HEX | FF |
| [same value] BIN | 11111111 |
| [same value] OCT | 377 |
| [same value] DEC | 255 |

*Press a base key to convert the shown value; press DEC to return to ordinary decimal.*

## 6-2 Bitwise logic — and, or, xor, Not

In base-n mode the SHIFT legends of the base keys give the logical operators and, or, xor and Not, which act bit by bit on the integer operands. Their precedence, from loosest to tightest, is: or, then xor, then and, then + and −, then × and ÷, then the unary Not. These words also work in decimal, where they force integer arithmetic.

**Example 18 — logic in hexadecimal / decimal**

| Operation | Display |
|---|---|
| [HEX] 1A and F EXE | A |
| [HEX] 1C or 3 EXE | 1F |
| [DEC] 12 and 8 EXE | 8 |
| [DEC] Not 0 EXE | -1 |

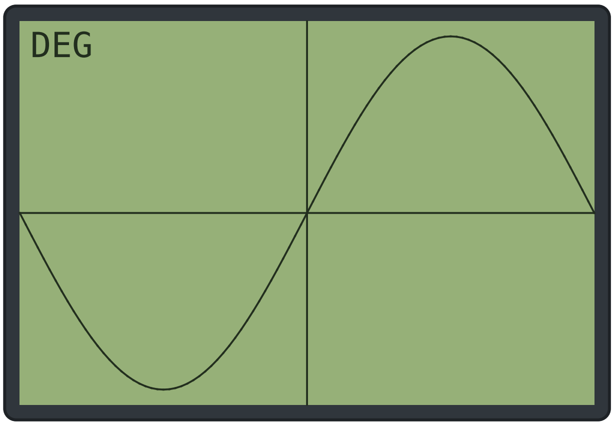
CHAPTER 7

# Graphing Functions

## 7-1 Drawing Y = f(X)

Enter an expression that uses the variable X (the X key), then press Graph. The calculator plots Y = f(X) across the display, drawing the axes with tick marks. Graph again with a new expression to overlay it on the previous curves — up to six curves are kept. Press Graph while a graph is showing to toggle back to the calculation screen; the curves are remembered, so pressing Graph once more brings them straight back. Pressing any ordinary key also returns you to the calculation screen.

**sin** → **X** → **Graph**




*A plot of Y = sin X with axes and tick marks.*

## 7-2 Setting the window — Range

Press SHIFT then √ (the Rng function) to open the Range editor. It lists six values that define the viewing window and the spacing of the tick marks. Move between fields with EXE (or the arrows), type a new value, and press Graph to redraw. The defaults are XMIN=-4.7, XMAX=4.7, XSCL=1, YMIN=-3.1, YMAX=3.1, YSCL=1.

- XMIN / XMAX — the left and right edges of the window.
- XSCL — the distance between tick marks along the x-axis.
- YMIN / YMAX — the bottom and top edges.
- YSCL — the tick spacing along the y-axis.

> **Note:** If XMAX is not greater than XMIN (or YMAX not greater than YMIN) the graph cannot be drawn and Ma ERROR is shown.

## 7-3 Built-in graphs — Bltin

Press SHIFT then Graph (the Bltin function) to choose a ready-made graph. Pick a number from the menu and the calculator plots it immediately:

- 1 sinX 2 cosX 3 tanX 4 $X^2$
- 5 X^3 6 √X 7 1/X 8 lnX

## 7-4 Tracing

While a graph is shown, press ◄ or ► to switch on the trace cursor. A pointer moves along the most recent curve, and the X and Y coordinates of the cursor are read out at the foot of the screen. Keep pressing the arrows to move the pointer one dot-column at a time.

CHAPTER 8

# Error Messages

When a calculation cannot be completed the display shows Ma ERROR. Press AC, DEL or any editing key to clear the message and correct the entry. The common causes are listed below.

| Cause | Example |
|---|---|
| Division by zero | 5÷0, or $x^{-1}$ of 0 |
| Argument outside a function's domain (e.g. log or √ of a non-positive number, $\sin^{-1}$ outside -1–1) | log0, √((-)4, $\sin^{-1}$(2 |
| Result too large to display (overflow, $\|x\| \geq 10^{100}$; base-n beyond about $9.2 \times 10^{18}$) | $10^{x}$200 |
| Factorial of a non-integer, a negative number, or a value above 69 | 3.5!, 70! |
| nPr / nCr with invalid arguments (negative, non-integer, or r > n) | 3nPr5 |
| Incomplete or malformed expression | 2+× |

# Appendix A — Key Reference

Every key with its primary function and its SHIFT (orange) and ALPHA (red) legends. Prefix keys (SHIFT, ALPHA, hyp) are one-shot.

| Key | Primary function | SHIFT / ALPHA |
|---|---|---|
| SHIFT | select orange legend | — |
| ALPHA | select red letter | — |
| hyp | sinh/cosh/tanh prefix | — |
| MODE | cycle DEG/RAD/GRA | SET (Norm/Fix/Sci setup) |
| DEL / AC | delete token / clear all | — |
| sin cos tan | trig functions | $\sin^{-1}$ $\cos^{-1}$ $\tan^{-1}$ · ALPHA L N O |
| ( ) | parentheses | )→ , (comma) · ALPHA P Q |
| log ln | base-10 / natural log | $10^x$ $e^x$ · ALPHA R S |
| $x^2$ $x^x$ | square / power | $^x\sqrt{}$ $x^{-1}$ · ALPHA T U |
| √ | square root | Rng (Range editor) · ALPHA V |
| x! | factorial | ALPHA W |
| Abs Int Frac | abs / integer / fraction | ″ ° ′ · ALPHA Y |
| X | graph variable X | → (store) |
| DEC HEX BIN OCT | select number base | and or xor Not |
| Graph | plot Y = f(X) | Bltin (built-in graphs) |
| 7 8 9 | digits | nPr nCr Ran# · ALPHA A B C |
| 4 5 6 | digits | Pol Rec % · ALPHA D E F |
| 1 2 3 | digits | ALPHA G H I |
| π EXP | pi / exponent (E) | e · ALPHA J K |
| 0 . (-) | digit / point / sign | ALPHA Z |
| Ans | last answer | — |
| M+ | add to memory M | M (recall) · ALPHA M |
| ◄► EXE | cursor / evaluate | — |

# Appendix B — Specifications

| Item | Specification |
|---|---|
| Display | Dot-matrix LCD, 96 × 64 pixels; 5×7 font, 16 characters × 8 rows |
| Entry line | Up to 32 characters over two rows, with a visible cursor |
| Number display | Norm (10 significant digits), Fix (0–9 decimals), Sci (1–10 sig. figures) |
| Overflow | Ma ERROR at $\lvert x\rvert \geq 10^{100}$ (about $9.2\times10^{18}$ in base-n) |
| Angle units | Degree, Radian, Gradient |
| Number bases | Decimal, Hexadecimal, Binary, Octal with and / or / xor / Not |
| Memories | Independent memory M; 26 variables A–Z; Ans; graph variable X |
| Functions | Trig & inverse, hyperbolic & inverse, log/ln, $10^x$/$e^x$, powers & roots, x!, %, nPr, nCr, Abs, Int, Frac, Pol, Rec, DMS |
| Graphing | Y = f(X) plots, up to 6 overlaid curves, 8 built-in graphs, Range editor, trace |
| Constants | π, e, Ran# (random), Ans |

# Index